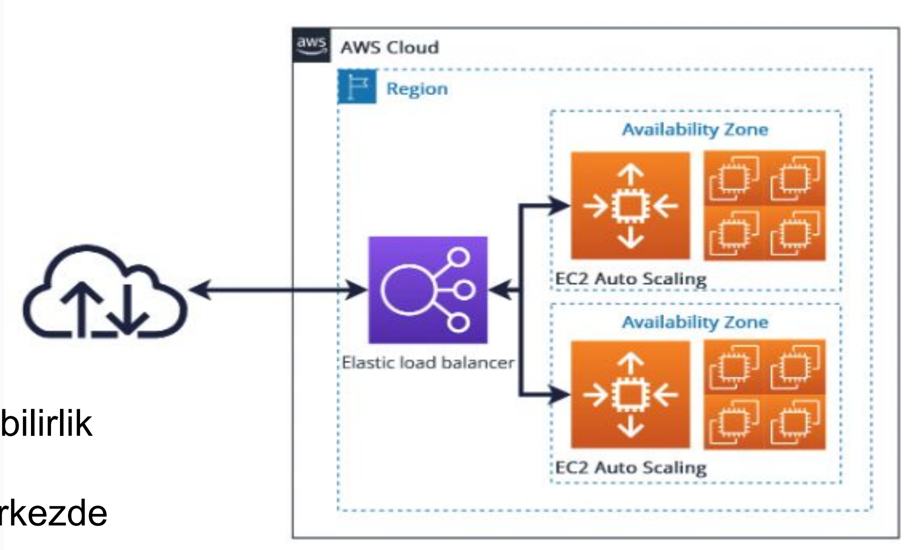
Sistem ve Network Atölyesi

High Availability And Scalability

High Availability

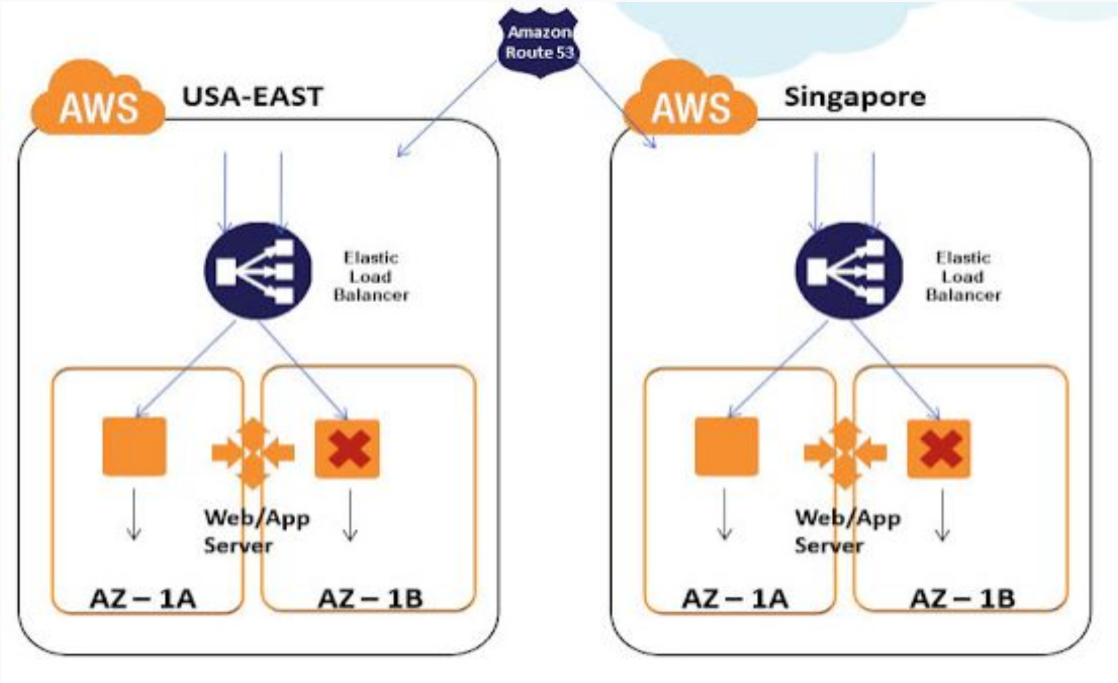
High Availability Nedir?

Bir nesnenin veya kaynağın yedekli kopyalarının oluşturulması ve biri başarısız olduğunda diğerinin devralmasıdır.

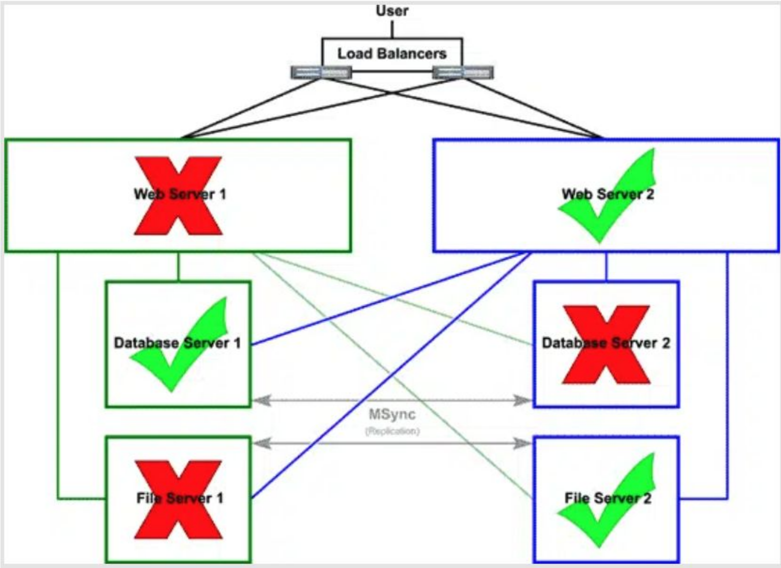


- AWS'de yüksek erişilebilirlik Multi-AZ (Çoklu Erişilebilirlik Alanı) dağıtımı ile sağlanır
- Veri merkezleri coğrafi olarak dağıtıldığı için bir merkezde yaşanan sorun diğerlerini etkilemez

High Availability and Fault Tolerance



Yüksek erişilebilirlik sistemi, çeşitli bileşenlerin arızalanması durumunda bile çalışmaya devam edebilir



High Availability and Fault Tolerance

‼️**AWS is Down** & let's understand How? 🤔

AWS Outage (US-EAST-1): Explained

Today, a major AWS outage affected many apps and services around the world. Here's a quick and easy explanation of what went wrong:
1️⃣ It all started with a DNS problem, that means AWS services couldn't find or connect to each other.
2️⃣ This caused DynamoDB to stop working. Many other AWS tools and customer applications depend on it, so they also started failing.
3️⃣ As a result, EC2 instances (virtual servers) and Lambda functions (serverless code) couldn't start or run properly.
4️⃣ AWS fixed the DNS issue, but some services are still recovering and syncing up.

This outage shows how even a small issue in one part of the cloud can affect many others, and how much of the internet depends on these systems. ⚙️

Follow live updates: https://lnkd.in/eDbfJTwV

#aws

AWS Health Dashboard

Updated less than 1 minute ago

Service health

View the current and historical status of all AWS services.

View your account health

Get a personalized view of events that affect your AWS account or organization.

Open your account health

Open and recent issues (1) | **Service history**

▼ Operational issue - Multiple services (N. Virginia)

Service
Multiple services

Severity
⚠ Degraded

RSS

Increased Error Rates and Latencies

Oct 20 11:22 AM PDT Our mitigations to resolve launch failures for new EC2 instances continue to progress and we are seeing increased launches of new EC2 instances and decreasing networking connectivity issues in the US-EAST-1 Region. We are also experiencing significant improvements to Lambda invocation errors, especially when creating new execution environments (including for Lambda@Edge invocations). We will provide an update by 12:00 PM PDT.

Oct 20 10:38 AM PDT Our mitigations to resolve launch failures for new EC2 instances are progressing and the internal subsystems of EC2 are now showing early signs of recovering in a few Availability Zones (AZs) in the US-EAST-1 Region. We are applying mitigations to the remaining AZs at which point we expect launch errors and network connectivity issues to subside. We will provide an update by 11:30 AM PDT.

Oct 20 10:03 AM PDT We continue to apply mitigation steps for network load balancer health and recovering connectivity for most AWS services. Lambda is experiencing function invocation errors because an internal subsystem was impacted by the network load balancer health checks. We are taking steps to recover this internal Lambda system. For EC2 launch instance failures, we are in the process of validating a fix and will deploy to the first AZ as soon as we have confidence we can

AWS OUTAGE (20 OCTOBER 2025)

Söz konusu kesinti, 1000'den fazla şirketi etkiledi.

Amazon & Amazon AWS

Pinterest

Snapchat

Roblox

Fortnite

Duolingo

Canva

Downdetector

HMRC

Vodafone

PlayStation

Pokémon Go.

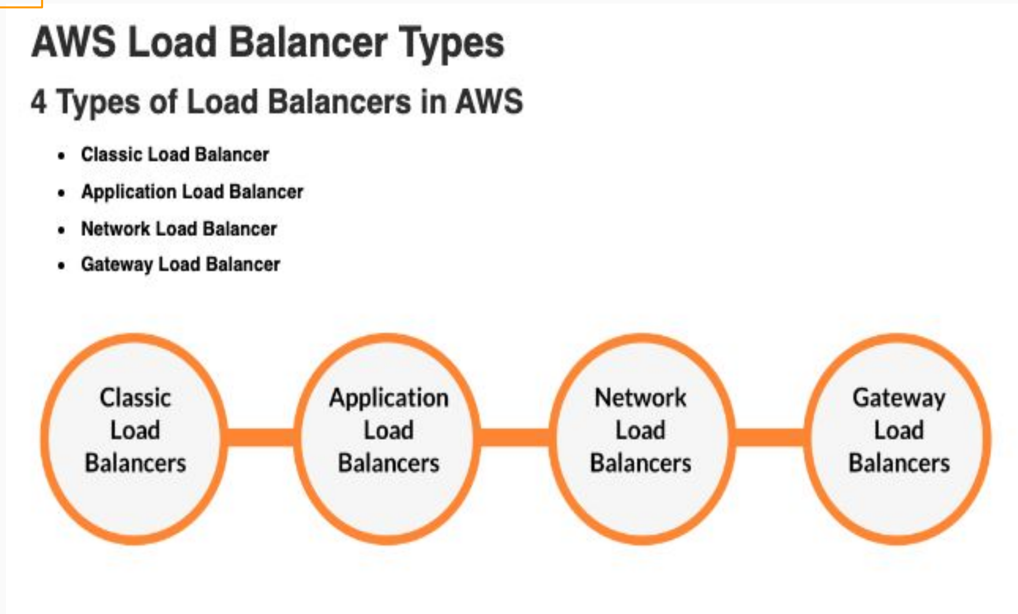
Coinbase

https://edition.cnn.com/business/live-news/amazon-tech-outage-10-20-25-intl

Elastic Load Balancing in AWS

AWS Elastic Load Balancing (ELB)

Elastic Load Balancing, AWS tarafından sunulan yönetilen bir yük dengeleme hizmetidir. Gelen trafiği EC2 örnekleri, konteynerler ve bir veya daha fazla Erişilebilirlik Alanı'ndaki IP adresleri gibi farklı kaynaklar arasında dağıtır.



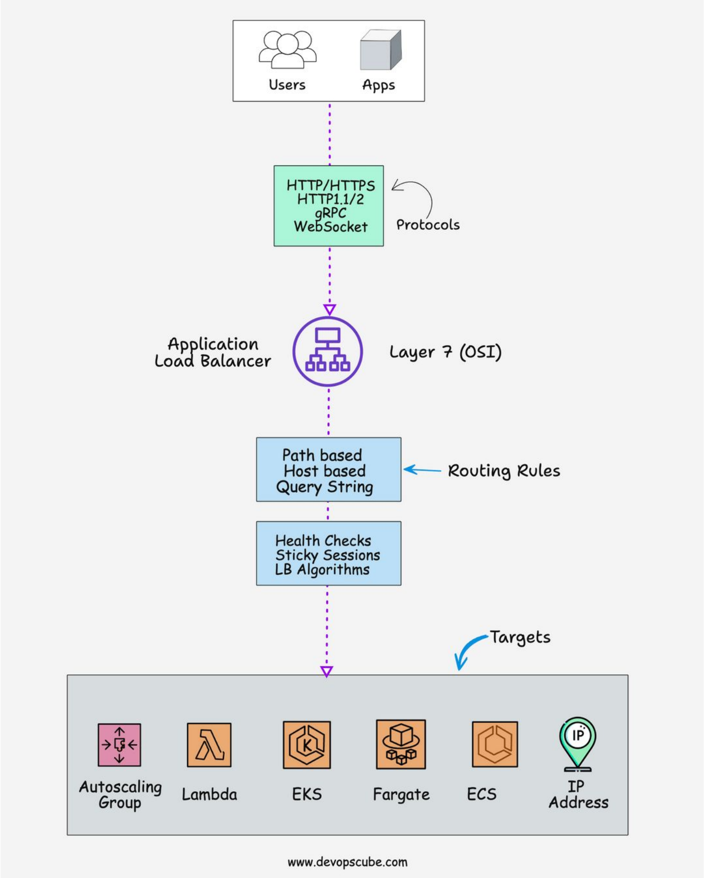
AWS Load Balancer Key Concepts

Target Groups: The primary work of a load balancer is to route traffic to backend instances or targets.A Target group is a logical grouping of targets such as ec2 instances, Lambda functions, EKS cluster, IP address, etc.
Listeners: The listener is a configuration where you specify the port and protocol for the front end of the load balancer and the protocol and port for the backends Meaning, the client has to interact with the load balancer on that specific port and protocol. It also forwards the incoming client requests to the configured backed target groups
Health Checks: The health check is the mechanism for the Load balancer to check the status of the backend targets and route only traffic to healthy targets.

Here is what you should know about health checks.

1. Monitor the health of registered targets using the **health check intervals** (default is 30 seconds)
2. Health check support HTTP, HTTPS, and TCP protocols.
3. NLB uses active and passive health checks
4. Health checks are part of target group configurations.

Application Load Balancer

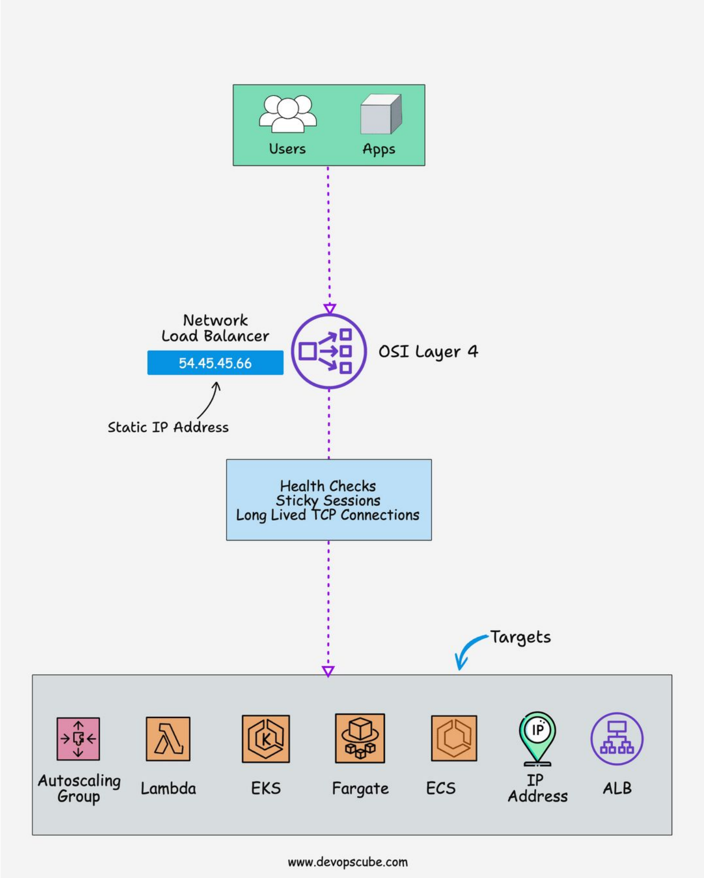


1. ALB runs at the application layer, which is **OSI layer 7**.
2. ALB supports routing rules like path-based routing, host-based routing, and query string routing.
3. You can add a security group to ALB to allow incoming traffic only on specific ports and sources.
4. ALB stickiness features allow requests from a particular client to be consistently routed to the same target.
5. Supports protocols like HTTP, HTTPS, HTTP/2, gRPC, WebSockets, etc.
6. Supports IPV4 and Dualstack IP address types (IPV4 & IPV6)
7. Targets can be Autoscaling Groups, Lambda Functions, Fargate, EKS cluster, ECS, or an IP address.
8. ALB does not support Static IP addresses.

Example Use Cases: Web applications, Microservices & API gateways

Source: devopscube.com

Network Load Balancer



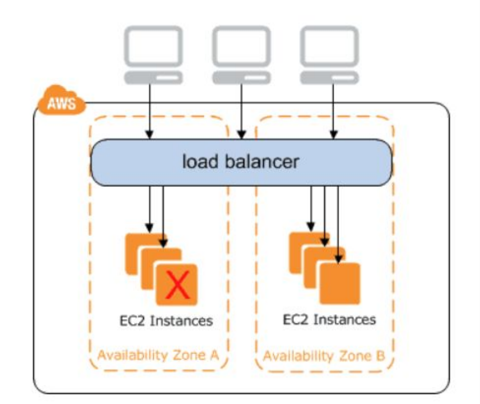
1. The network load balancer is an ideal choice for high-performance applications. It is built for handling volatile traffic patterns and **millions of requests** per second.
2. Support protocols like TCP and UDP
3. Support static IP address.
4. You cannot add a security group to NLB.
5. NLB runs at the transport layer which is **OSI layer 4**
6. With NLB, you can route requests to multiple applications running on a single ec2 instance on different ports.
7. NLB can handle more traffic than ALB.

Example Use Cases: Gaming/VoIP/streaming media servers, financial trading applications

Source: devopscube.com

Classic Load Balancer

Real-World Use Cases: Firewall Management, intrusion detection, and prevention systems.



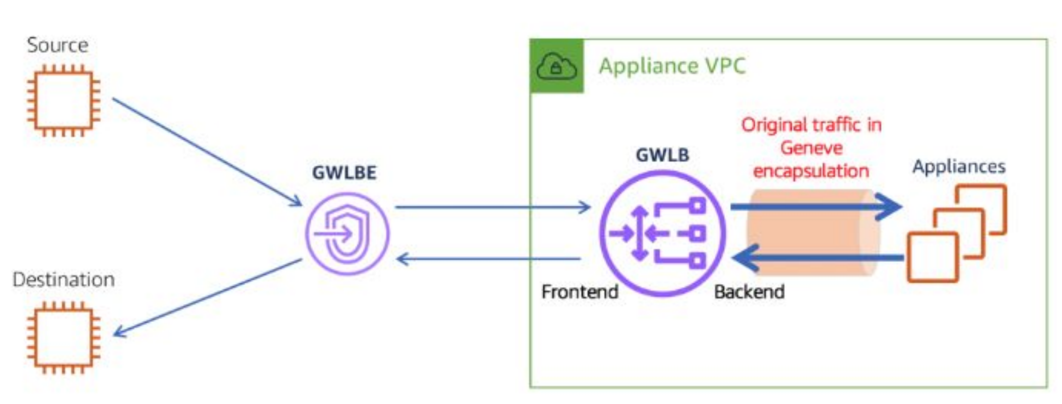
- The Load Balancer which balances the traffic across multiple instances in multiple availability zones is called a Classic Load Balancer.
- It supports both EC2 Classic EC2-VPC and Increases the availability of your application by sending traffic to healthy Instances.
- Supports HTTP, HTTPS, TCP, and SSL listeners and supports sticky sessions using application-generated cookies.
- To make sure that the instances you have registered can handle the demand Keep roughly the same number of instances registered with the load balancer in each Availability Zones.

Regional Limit	
LB per region	20
Load Balancer Components Limit	
Listeners	50
Security groups	5
Registered instances	1000
Subnets per Availability Zone	1

Source: https://k21academy.com/

Gateway Load Balancer

Real-World Use Cases: Firewall Management, intrusion detection, and prevention systems.



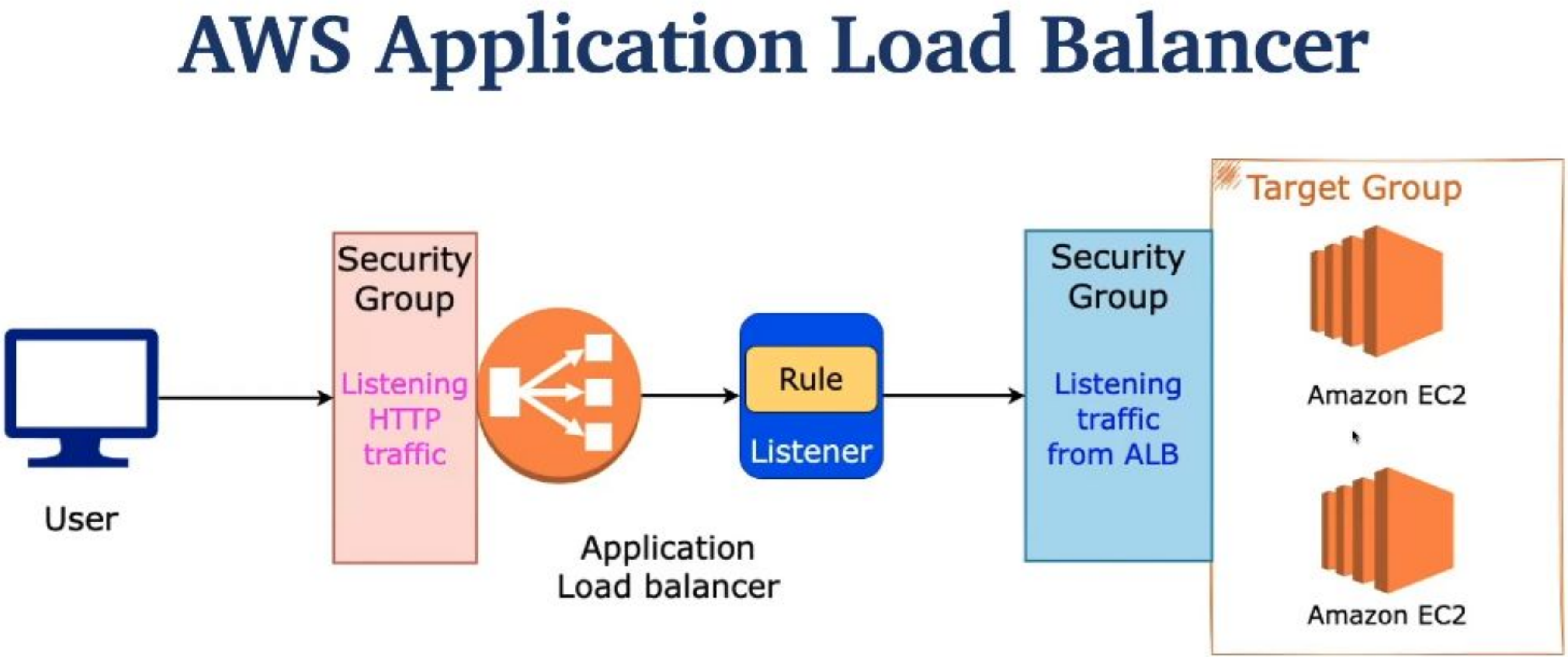
1. The gateway load balancer is used for creating, scaling, and managing virtual services like intrusion detection in the AWS cloud.
2. GLB runs at the network layer, that is **OSI layer 3**.
3. It can load balancer traffic across multiple **virtual appliances** (Firewalls, WAFs, CDNs, etc)
4. It can act as a single entry and exit point for all traffic.
5. Forwards traffic using GENEVE protocol on port 6081

Regional Limit	
LB Per Region	20
Load Balancer Components Limit	
Gateway Load Balancers per VPC	10
Target groups with GENEVE protocol	100
Targets per Availability Zone per target group with GENEVE protocol	300

ELB

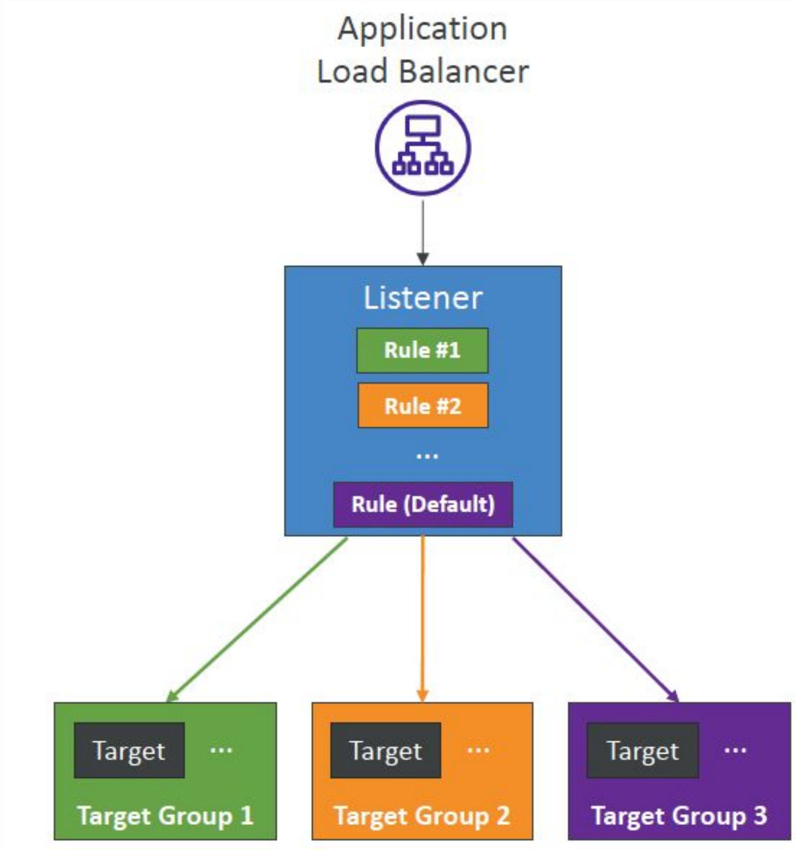
Feature	Application Load Balancer	Network Load Balancer	Classic Load Balancer
Protocols	HTTP, HTTPS	TCP	TCP, SSL/TLS, HTTP, HTTPS
Platforms	VPC	VPC	EC2-Classic, VPC
Health checks	✓	✓	✓
CloudWatch metrics	✓	✓	✓
Logging	✓	✓	✓
Path-Based Routing	✓		
Host-Based Routing	✓		
Native HTTP/2	✓		
			✓
SSL offloading	✓		✓
Static IP		✓	
Elastic IP address		✓	
Slow start	✓		

ALB


AWS Application Load Balancer
Target Group
Security
Group
Listening
HTTP
traffic
Rule
Listener
Security
Group
Listening
traffic
from ALB
Amazon EC2
Amazon EC2
User
Application
Load balancer

ALB- Listener Rules

- Processed in order (with Default Rule)
- Supported Actions (forward, redirect,
- fixed-response)
- Rule Conditions:
  - host-header
  - http-request-method
  - path-pattern
  - source-ip
  - http-header
  - query-string



EC2 Auto Scaling

AutoScaling Group(ASG)

EC2 instance'ları otomatik ekleyen/çıkaran group
Bileşenler:

- Launch Template
- Desired Capacity (istediğin minimum/maximum instance sayısı)
- Scaling Policies (ölçekleme kuralları)

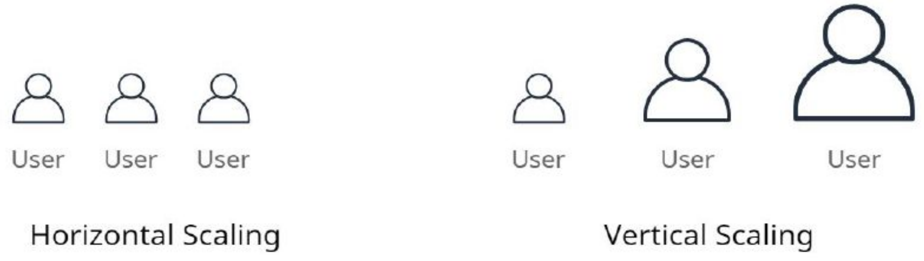
Faydalar:

- Trafik arttığında yeni instance eklenir
- Trafik azalınca gereksiz instance'lar kapanır
- Maliyet optimizasyonu + yüksek erişilebilirlik

we can configure automatic scaling for all of the scalable resources :

- **Amazon EC2**
- **Amazon EC2 Spot Fleets**
- **Amazon ECS**
- **Amazon DynamoDB**
- **Amazon Aurora**

It scales across multiple Availability Zones within the same AWS region.

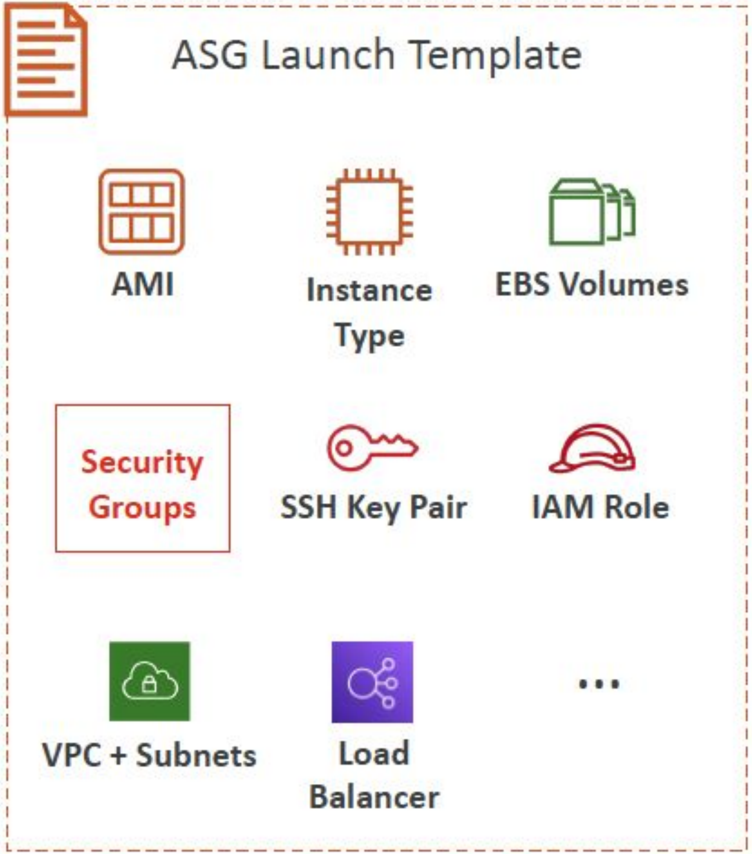


Auto Scaling Group Attributes

AWS Auto Scaling Components

1- Launch Configuration (Template)
- AMI [AWS AMIs or Custom AMIs]
- Instance Type
- Storage
- SG
- KEY Pair

2- Autoscaling Group
- Launch Config
- Network/Subnets
- Scaling Up policies for increase
- Scaling Down policies for decrease
- Monitoring & Alarms



EC2 Auto Scaling

The ways to scale Auto Scaling Groups are as follows:

Manual Scaling
Update the desired capacity of the Auto Scaling Group manually.

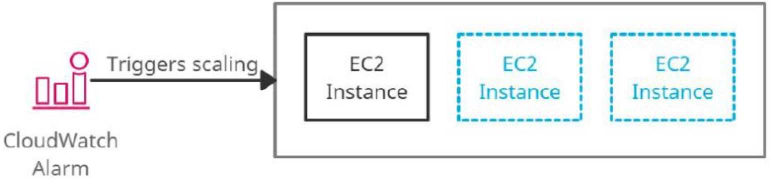
Scheduled Scaling
This scaling policy adds or removes instances based on predictable traffic patterns of the application.

Example:
Scale-out on every Tuesday or Scale in on every Saturday

Dynamic Scaling

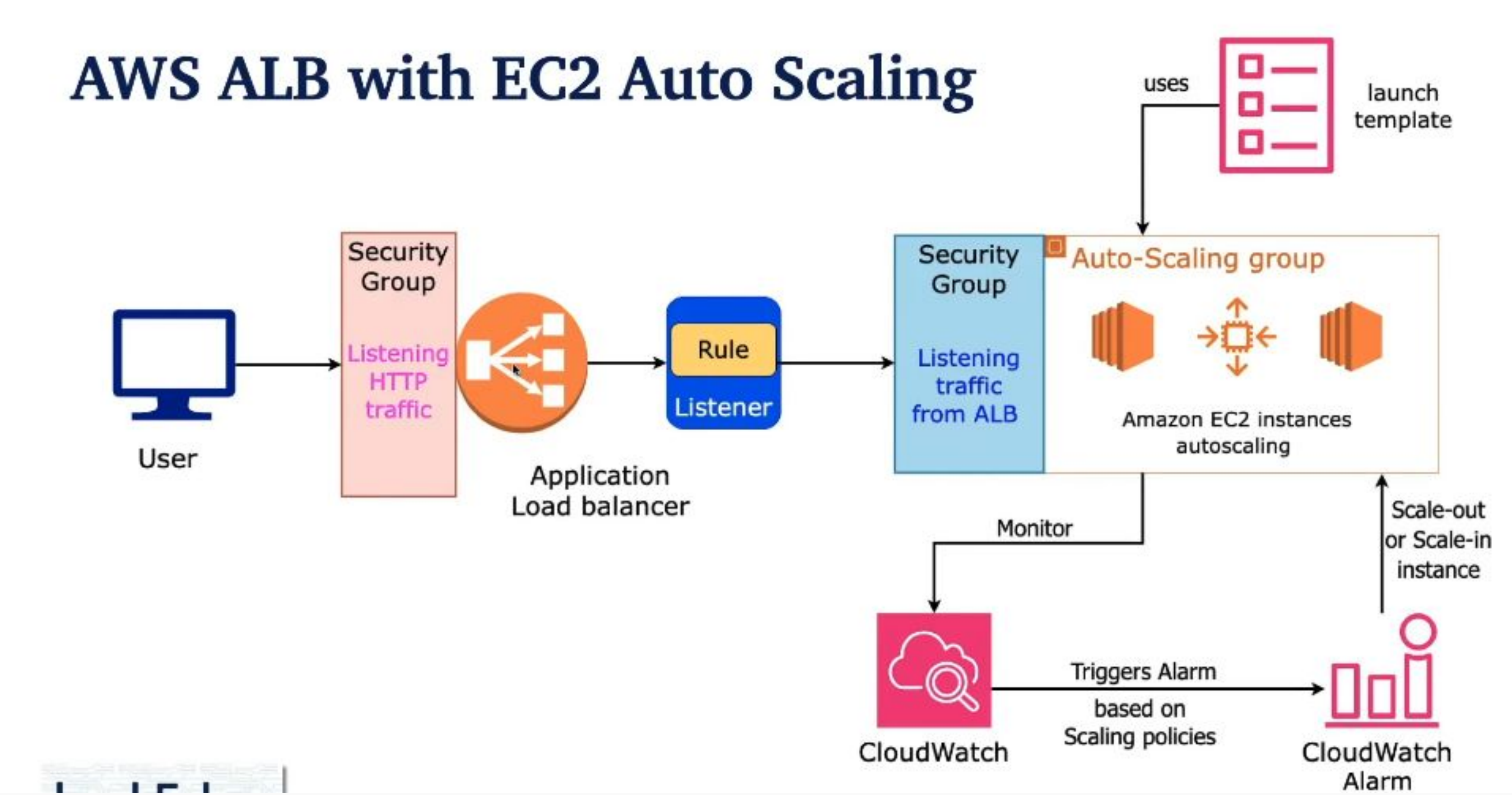
- *Target tracking scaling policy*: This scaling policy adds or removes instances to keep the scaling metric close to the specified target value.
- *Simple Scaling Policy:* This scaling policy adds or removes instances when the scaling metric value exceeds the threshold value.
- *Step Scaling Policy:* This scaling policy adds or removes instances based on step adjustments (lower bound and upper bound of the metric value).

- The Cooldown period is the time during which an Auto Scaling group doesn't launch or terminate any instances before the previous scaling activity completes.



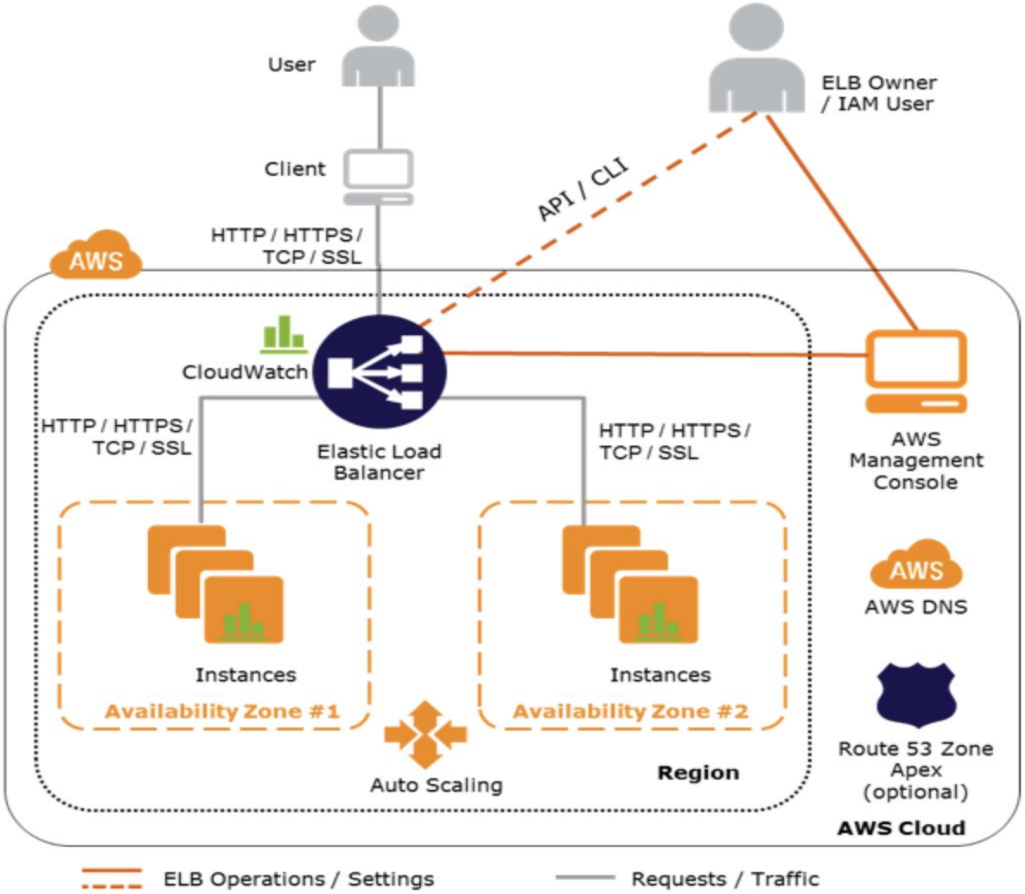
Amazon EC2 Auto Scaling using CloudWatch Alarm

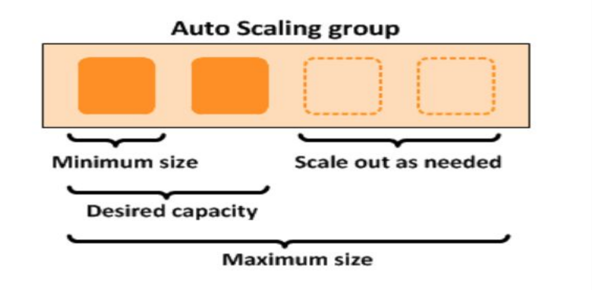
ALB



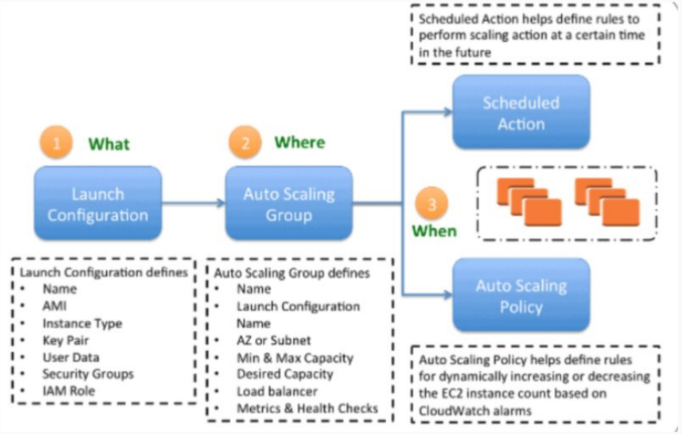
AWS ALB with EC2 Auto Scaling
uses
launch template
Security Group
Listening HTTP traffic
User
Application Load balancer
Rule
Listener
Security Group
Listening traffic from ALB
Auto-Scaling group
Amazon EC2 instances autoscaling
Monitor
Scale-out or Scale-in instance
CloudWatch
Triggers Alarm
based on Scaling policies
CloudWatch Alarm

EC2 Auto Scaling





Auto Scaling-Configuration



CloudWatch & SNS

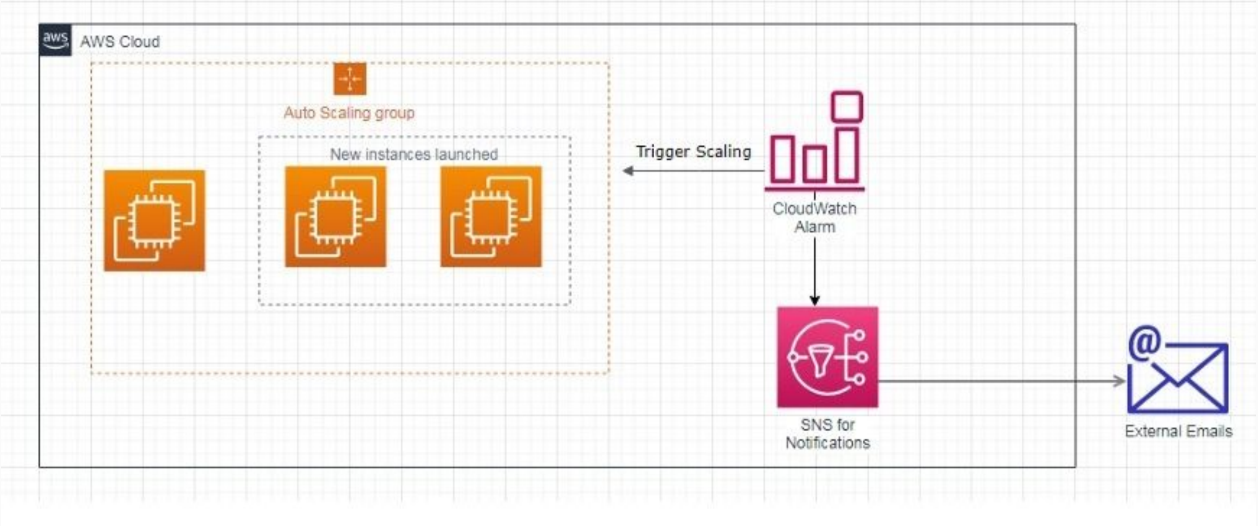
CloudWatch: AWS monitoring servisi

- CPU, bellek, disk, ağ kullanımı ölçümü
- Custom metric eklenebilir

SNS (Simple Notification Service):

- CloudWatch alarm → SNS topic → e-posta/SMS/Slack bildirimi

Örnek: CPU %80 → ASG scale-out + admin’e e-posta



Multi-AZ Auto Scaling Web App

Multi-AZ Auto Scaling Web App

- Launch Template oluştur (Amazon Linux + Apache)
- ASG kur (Min:2, Max:5, Desired:2)
- ALB ekle ve ASG’yi target group’a bağla
- CloudWatch CPU alarm kur (örn. %70 scale-out, %30 scale-in)
- SNS notification ile e-posta bildirimi ekle
- Load test → yeni instance otomatik ekleniyor mu kontrol et